Solutions to JEE Main - 9 | JEE-2022

PHYSICS

SECTION-1

1.(C) Let A and B carry charge Q and –Q. When C is brought in contact with A, the new charges in A and C become Q/2. Now when C is brought in contact with B, the new charge in both B and C become $\frac{-Q}{4}$.

$\therefore \quad F_{old} = \frac{KQ^2}{a^2} \qquad F_{new} = \frac{K(Q/2)(Q/4)}{a^2} = \frac{F_{old}}{8}$

2.(C) $E_B = \frac{KQ}{(3R)^2} = \frac{KQ}{9R^2}$

$V_A = \frac{KQ}{R} \quad ; \quad \frac{V_A}{E_B} = \frac{KQ}{R}\frac{9R^2}{KQ} = 9R$

3.(C) $E = \frac{-d}{dx}(9x^3+3) = -27x^2$

$E\,(-3,2,0) = -243\ V/m$

4.(A) Potential due to $\text{Ring}^1 = \frac{KQ}{R}$

Potential due to $\text{Ring}^2 = \frac{KQ}{R}$

Total Potential at centre $= \frac{Q}{2\pi\varepsilon_0 R}$

5.(A) There are 18.6×10^{18} electrons in 3C charge

6.(C) The potential an equatorial line of dipole is always zero.

7.(B) The charge will distribute as they are at different potential

Let Q_1 and Q_2 be charges at body 1 and 2 respectively

$Q_1 + Q_2 = 2q - q$ {Conservation of charge}

$\frac{KQ_1}{r} = \frac{KQ_2}{3r}$

$Q_2 = 3Q_1 \quad ; \quad 4Q_1 = q$

$Q_1 = \frac{q}{4} \quad ; \quad Q_2 = \frac{3q}{4}$

8.(B) $E = 100\ N/C$

$\theta = 60°$

$|\overline{p}| = 10^{-10}\,C-m$

$U = -\overline{p}\cdot\overline{E} = -|\overline{p}||\overline{E}|\cos 60°$

$= -10^{-10} \times 100 \times \frac{1}{2} \quad ; \quad = -5\times10^{-9}\,J$

9.(A) Potential always decreases in the direction of field lines.

10.(A)

11.(C) Electric field at any $x = \dfrac{KQx}{(R^2 + x^2)^{3/2}}$

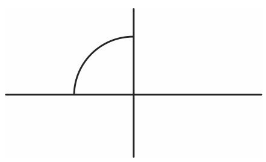
Force on the charge $q = \dfrac{-KQqx}{(R^2 + x^2)^{3/2}}$; $= \dfrac{-KQq}{R^3} x \; \{R >>> x\}$

Acceleration of charge $q = \dfrac{-KQq}{mR^3} x$

This is an SHM

$\omega = \sqrt{\dfrac{KQq}{mR^3}}$; $T = 2\pi\sqrt{\dfrac{mR^3}{KQq}}$

12.(C)



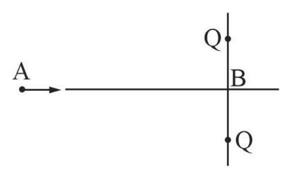
$\lambda = \dfrac{Q/2}{\dfrac{\pi(2R)}{2}} = \dfrac{Q}{2\pi R}$

$\overline{E}$ **.** field due to Quarter Ring is $\dfrac{K\lambda}{R}$ in both $\hat{i}$ and $-\hat{j}$ directions.

$\therefore \quad \dfrac{Q}{8\pi^2\varepsilon_0 R}\hat{i} - \dfrac{Q}{8\pi^2\varepsilon_0 R}\hat{j}$

13.(D) The graph D corresponds to the potential of a point in a uniformly non conducting sphere.

14.(A) Using conservation of energy between points A and B



$\dfrac{1}{2}mv^2 + 0 = \dfrac{2KQq}{a} + 0$

$v = \left(\dfrac{4KQq}{am}\right)^{1/2}$

15.(A) Potential due to charge at centre $= \dfrac{K}{R/3}\left(\dfrac{q}{2}\right)$

$= \dfrac{3Kq}{2R}$

Potential due to charge of shell $= \dfrac{Kq}{R}$ $\quad \therefore$ Total potential $= \dfrac{5Kq}{2R} = \dfrac{5q}{8\pi\varepsilon_0 R}$

16.(D) Dipole moment, $\overline{p} = qd$

Electric field, $\overline{E} = E$

angle between them = 30°

Net force = 0

Net force on dipole in uniform E field is zero.

17.(A) Flux through curved surface area is zero as angle Between E.Field and area is 90°.

For area 1, $x = 1m$

$E = 50\hat{i}$

$A = -25\times 10^{-4}\hat{i}$

$\phi_1 = \overline{E}.\overline{A} = -125\times 10^{-3}$

$= -1.25\times 10^{-1}$

$= -.125 Nm^2/C$

For area 3, $\overline{E} = 100\hat{i}$ $(\because x = 2m)$

$\overline{A} = 25\times 10^{-4}\hat{i}$

$\phi_3 = 25\times 10^{-2} = .25\ Nm^2/C$

$\therefore$ total flux $= .125 Nm^2/C$

18.(D) $\int_{V_0}^{V_A} dV = -\int_0 E'.dr$; $V_A - V_0 = -\int_0^1 20y^3 dy$

$V_A - V_0 = -\left.\frac{20y^4}{4}\right]_0^1$; $= -5(1-0) = -5V$

19.(B) $E_A = \frac{\sigma}{2\varepsilon_0} + \frac{2\sigma}{2\varepsilon_0} = \frac{3\sigma}{2\varepsilon_0}$

$E_B = \frac{-2\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{-\sigma}{2\varepsilon_0}$; $\frac{E_A}{E_B} = +3$

20.(A) Force $= \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{d^2} = F$

Force in medium $= \frac{1}{4\pi\varepsilon_0 K}\frac{q_1q_2}{(2d)^2} = \frac{F}{12}$

SECTION-2

21.(32) Force on the charge 2q will be because of the other 3 charges.

$F_1 = \frac{K(2q)(2q)}{a^2}\downarrow$

$F_2 = \frac{K(2q)(q)}{a^2}\rightarrow$

$F_3 = \frac{K(2q)(q)}{a^2}\rightarrow$

Using principle of superposition

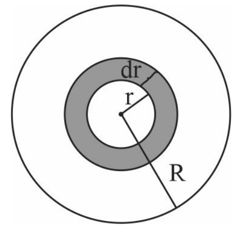
$\overline{F}_{total} = \overline{F}_1 + \overline{F}_2 + \overline{F}_3$

$= \frac{K(4q^2)}{a^2}(-\hat{j}) + \frac{K(4q^2)}{a^2}(\hat{i})$

$= \frac{K4q^2}{a^2}(\sqrt{2}) = \frac{Kq^2}{a^2}\sqrt{32}$

22.(45) $\rho(r) = \rho_0 r^2$

Let us take a shell at radius r and a width of dr



Volume of shell, $dV = 4\pi r^2 dr$

Charge on shell, $dq = 4\pi r^2 . \rho_0 r^2 dr$

$= 4\pi\rho_0 r^4 dr$

Total charge on sphere, $Q = \int dQ$

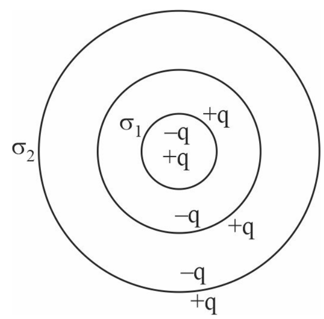
$= \int_0^R 4\pi\rho_0 r^4 dr$

$= \dfrac{4\pi\rho_0 R^5}{5}$

Now making a Gaussian surface at radius 3R, we get $\oint E \cdot dA = \dfrac{Q_{enc}}{\varepsilon_0}$

$E \cdot \cancel{4\pi}(3R)^2 = \dfrac{\cancel{4\pi}}{5}\dfrac{\rho_0 R^5}{\varepsilon_0}$; $E = \dfrac{\rho_0}{45}\dfrac{R^3}{\varepsilon_0}$

23.(9) q is kept at centre of shells. So charges will be induced at all the surface of the shells.



$\sigma_1 = \dfrac{q}{4\pi R^2}$

$\sigma_2 = \dfrac{q}{4\pi(3R)^2}$

$\dfrac{\sigma_1}{\sigma_2} = 9$

24.(1) $\phi_1 = 0$

$\phi_2 = \dfrac{q}{24\varepsilon_0}$

$\phi_3 = \dfrac{q}{24\varepsilon_0}$

$\dfrac{\phi_2 - \phi_1}{\phi_3 - \phi_1} = 1$

25.(8) Potential difference between these two points = 2V.

$\therefore \quad 4n = 8$

Chemistry

SECTION-1

1.(D) A atoms occupy corners & face centre of cubic unit cell.

$\therefore$ No. of A atom = $8\times\frac{1}{8}+6\times\frac{1}{2}=4$

B atoms occupy octahedral voids

$\therefore$ no. of B atoms = 4

Formulae of compound A_4B_4

If all face centred atoms along the one axis is removed, then

No. of A atoms remained $=8\times\frac{1}{8}+4\times\frac{1}{2}=3$

$\therefore$ formulae of solid $= A_3B_4$

2.(D) In FCC lattice,

$a\sqrt{2}=4r$

$r=\frac{a\sqrt{2}}{4}=\frac{1.414\times 3.16}{4}=1.117\text{Å}$

For octahedral void,

$2(r+R)=a$

$2R=a-2r=0.926\text{Å}$

3.(A) In DC, C atoms occupy ccp & half of tetrahedral voids packing efficiency of DC is 34%

4.(A) We know that

$\text{Density}=\frac{Z\times M}{N_A\times a^3}$

$=\frac{4\times 58.5}{6.023\times 10^{23}\times(0.5627)^3\times 10^{-21}}=2.1805\,g/cm^3$

Observed density is 2.164 g/cm^3 which is less then calculated density because some places are missing. Actual constituent units per unit cell can be calculate as:

$Z=\frac{(0.5627\times 10^{-7})^3\times 2.164\times 6.023\times 10^{23}}{58.5}=3.969$

Missing units = 4 – 3.969 = 0.031

% missing = $\frac{0.031}{4}\times 100=0.775\%$

5.(B) Frenkel defect is generally available for smaller size of cations.

6.(B) 'A' constitute ccp lattice

$\therefore$ no. of A ion present = 4

B & C occupies alternate T_d voids

Along one body diagonal,

no. of A atoms removed $=2\times\frac{1}{8}=\frac{1}{4}$

no. of B atoms removed = 1

no. of C atoms removed = 1
formulae of compound

$A_{4-\frac{1}{4}}B_{4-1}C_{4-1}$

$A_{3.75}B_3C_3$

7.(C) $\frac{r^+}{r^-} = 0.414 - 0.732$, Co. No. –6

$\frac{r^+}{r^-} = 0.225 - 0.414$, Co. No – 4

Hence (C)

8.(C) $ZnO \xrightarrow{\Delta} Zn^{+2} + \frac{1}{2}O_2 + 2e^-$

Trapping of e^- at the site vacated by oxide ion, cause yellow colour & conducting nature of ZnO.

9.(A) Radius ratio $\frac{r^+}{r^-} = \frac{88}{200} = 0.44$. This corresponds to co-ordination number 6.

10.(D) Schottky defects decreases density of crystal.
Schottky defects is usually observed when size of cation & anion are almost same
Schottky defects are thermodynamic intrinsic defects.

11.(B) no. of Q^{2-} ions = 6

No. of T_d voids occupied by $P^{x+} = 12 \times \frac{1}{2} = 6$

For stable ionic lattice, charge on cation should be equal to charge on anion

$2 \times 6 = 6 \times x \quad \therefore \quad x = 2$

12.(C) Let Cu^{+2} be x

then $Cu^{+1} = (1.92 - x)$

total charge on cation = total charge on anion

$\therefore \quad 2 \times x + 1 \times (1.92 - x) = 1 \times 2$

$x = 0.08$

$\therefore \quad Cu^{2+} = 0.08$

$Cu^+ = 1.92 - 0.08 = 1.84$

Molar ratio of $Cu^+ \,\&\, Cu^{+2} = \frac{0.08}{1.84} = 1:23$

13.(D) Percentage of void space in a FCC lattice
= 100 – 74 = 26%

$\therefore \quad$ void space per unit cell $= \frac{26}{100} \times$ Volume of unit cell

$= \frac{26}{100} \times 4 \times 4 \times 4$

$= 16.64 Å^3$

14.(A) $\text{density(d)} = \dfrac{Z \times M}{N_A \times a^3}$ (1)

$\dfrac{\text{packing fraction}}{\text{density}} = 0.1\,cm^3/gm$

Packing fraction = $\dfrac{\frac{4}{3}\pi r^3 \times z}{a^3}$

$\therefore \quad \dfrac{z \times \frac{4}{3}\pi r^3}{a^3 \times d} = 0.1 \qquad \therefore \quad d = \dfrac{\frac{4}{3}\pi r^3 \times z}{a^3 \times .0.1}$ (2)

From (1) and (2) we have $\dfrac{Z \times M}{N_A \times a^3} = \dfrac{\frac{4}{3}\pi r^3}{a^3 \times 0.1}$

$\therefore \quad M = \dfrac{\frac{4}{3}\pi r^3 \times N_A}{0.1} = \dfrac{\frac{4}{3}\pi \times (10^{-8})^3 \times 6 \times 10^{23}}{0.1} = 8\pi$

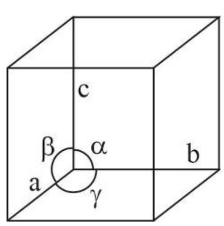
15.(B) Semi conductor is $Y_2Ba_2Cu_3O_7$
For one mole of semi conductor,
Mole of Y required = 2
Moles of Ba required = 2
Moles of Cu required = 3
$\therefore$ molar ratio of Y_2O_3, BaO_2 & CuO
Should be 1 : 2 : 3

16.(C) In FCC lattice, the closest distance is 2r
In FCC,

$\sqrt{2}a = 4r \;;\; 2r = \dfrac{\sqrt{2}a}{2} = \dfrac{a}{\sqrt{2}} = \dfrac{4.07}{\sqrt{2}} = 2.877\text{Å}$

17.(B)

18.(B)



$\alpha = \beta = \gamma = 90°$
$a = b \neq c$

19.(D) Na_2O (Antifluorite structure)

$O^{2-} \rightarrow$ ccp arrangement

$Na^+ \rightarrow$ All tetrahedral voids

$\therefore$ % of tetrahedral voids occupied $= \dfrac{8}{8} \times 100 = 100\%$

20.(C)

SECTION-2

21.(6) No. of atoms present in 78 gm of potassium $= \frac{78}{39} \times N_A$

$= 2N_A$

No. of atoms present in BCC = 2

$\therefore$ no. of unit cells $= \frac{2 \times 6 \times 10^{23}}{2} = 6 \times 10^3$

22.(3) density $= \frac{\text{mass}}{\text{volume}}$; volume $= \frac{\text{mass}}{\text{density}}$

$a^3 = \frac{58.5}{2.167}$; $a = \left(\frac{58.5}{2.167}\right)^{\frac{1}{3}} = 3$

23.(5) for a simple cubic unit cell,

a = 2r, where r is the radius of element if the radius of the larger atom that can be placed = R

Then 2R = 366 pm

$\therefore$ $2r + 2R = \sqrt{3}a$

$a + 366 = \sqrt{3}a$

$a = \frac{366}{(\sqrt{3}-1)} = \frac{366}{0.732} = 500\,\text{pm}$

Now $d = \frac{Z \times M}{N_A \times a^3}$

$d = \frac{1 \times 125}{6 \times 10^{23} \times (5 \times 10^{-8})^3}$; $d = \frac{10}{6} = \frac{5}{3}$ Value of 3d = $\frac{5}{3} \times 3 = 5$

24.(289.86) In NaCl type crystal,

Cl^- occupies cubic close packing

Na^+ present in octahedral voids

$\frac{r^+}{r^-} = 0.414$; $r^- = \frac{r^+}{0.414} = \frac{120}{0.414} = 289.86$

25.(12) density $= 10\,g/cm^3$

Mass = 100 gm

Volume $= \frac{100}{10} = 10\text{cm}^3$

Volume of one unit cell = $a^3 = (2 \times 10^{-8})^3 = 8 \times 10^{-24}\,\text{cm}^3$

No. of unit cell $= \frac{\text{Total volume}}{\text{volume of one unit cell}} = \frac{10}{8 \times 10^{-24}} = \frac{10^{25}}{8}$

But one unit cell contains 4 atoms,

So total number of atoms $4 \times \frac{1}{8} \times 10^{25} = 5 \times 10^{24}$

Now $\frac{xy}{10} = \frac{5 \times 24}{10} = 12$

Mathematics

SECTION-1

1.(A) $\left[x+\frac{1}{2}\right]>0$ and $\left[x+\frac{1}{2}\right]\neq 1$

$\therefore \quad x+\frac{1}{2}\geq 2$ and $2x^2+x-1>0$

$\Rightarrow \quad x\geq \frac{3}{2}$ and $(2x-1)(x+1)>0 \qquad \therefore \quad x\in\left[\frac{3}{2},\infty\right)$

2.(A) $f(f(x))=\dfrac{3\left(\frac{3x+5}{2x-3}\right)+5}{2\left(\frac{3x+5}{2x-3}\right)-3}=x$

$\therefore \quad f(f(x))=x \Rightarrow f_2(x)=x$

$\Rightarrow \quad f_3(x)=f(x)$

$\Rightarrow \quad f_4(x)=x$; $f_{2008}(x)+f_{2009}(x)=x+\frac{3x+5}{2x-3}=\frac{2x^2+5}{2x-3}$

3.(C) Domain $=(-2,1) \Rightarrow 0<\frac{\sqrt{4-x^2}}{1-x}<\infty \quad \Rightarrow -\infty<\log\frac{\sqrt{4-x^2}}{1-x}<\infty$

4.(B) $\ln|\ln|x||\geq 0 \qquad\qquad |x|^2-7|x|+10\leq 0$

$\Rightarrow \quad |\ln|x||\geq 1 \qquad\qquad \Rightarrow \quad (|x|-2)(|x|-5)\leq 0$

$\Rightarrow \quad \ln|x|\in(-\infty,-1]\cup[1,\infty) \qquad\qquad \Rightarrow \quad 2\leq|x|\leq 5$

$\Rightarrow \quad |x|\in\left(0,\frac{1}{e}\right]\cup[e,\infty) \qquad\qquad \Rightarrow \quad x\in[-5,-2]\cup[2,5]$

$\Rightarrow \quad x\in(-\infty,-e]\cup\left[-\frac{1}{e},0\right)\cup\left(0,\frac{1}{e}\right]\cup[e,\infty) \qquad \therefore \quad x\in[-5,-e]\cup[e,5]$

5.(B) $f(f(x))=\frac{1}{\sqrt{1+2x^2}},\ f(f(f(x)))=\frac{1}{\sqrt{1+3x^2}}....$

6.(B) $\frac{\sin x+\sin 7x}{\cos x+\cos 7x}=\tan 4x$

7.(D) $a=\lim\limits_{x\to 0}\frac{\ln(1+(\cos 2x-1))}{\cos 2x-1}\times\frac{\cos 2x-1}{3x^2}=-\frac{2}{3}$

$b=\lim\limits_{x\to 0}\left(\frac{\sin^2 2x}{4x^2}\right)\times\frac{4x^2}{x^2\left(\frac{1-e^x}{x}\right)}=-4$

$c=\lim\limits_{x\to 1}\frac{\frac{1}{2\sqrt{x}}-1}{\frac{1}{x}}=-\frac{1}{2}$

8.(A) $\lim_{x \to 2^+} [5-2x] = 0$

$\lim_{x \to 2^-} [|x-2| + a^2 - 6a + 9] = [(a-3)^2] = 0$

$\therefore \quad (a-3)^2 < 1 \Rightarrow 2 < a < 4$

9.(C) $\lim_{x \to 0^+} f(x) = \lim_{n \to \infty} (-1)^n = \pm 1$

If $x = \dfrac{1}{2^{2^n}}$ then $2x \neq \dfrac{1}{2^{2^n}} \Rightarrow f(2x) = 0$

10.(C) $f(0^-) = e^a,\, f(0) = b, c = 1$

$f(0^+) = \lim_{x \to 0^+} \dfrac{(x+1)^{1/3} - 1}{(x+1)^{1/2} - 1} = \dfrac{2}{3}$

11.(D) $\lim_{x \to 0} \dfrac{ae^{\sin x} + be^{-\sin x} - c}{x^2} .... [a + b - c = 0]$

$\lim_{x \to 0} \dfrac{\cos x(ae^{\sin x} - be^{-\sin x})}{2x} ... [a - b = 0]$

$= \lim_{x \to 0} \dfrac{-\sin x(ae^{\sin x} - be^{-\sin x}) + \cos^2 x(ae^{\sin x} + be^{-\sin x})}{2}$

$= \dfrac{a+b}{2}$

$$\therefore \quad \left.\begin{matrix} a + b = 4 \\ a = b \\ a + b - c \end{matrix}\right] \begin{matrix} a = b = 2 \\ c = 4 \end{matrix}$$

12.(D)

13.(B)

14.(B)

15.(D) $f\left(\dfrac{3}{2}\right) = \dfrac{9}{4}$

$f\left(\dfrac{9}{4}\right) = \dfrac{9}{2} - 3 = \dfrac{3}{2}$

$f\left(\dfrac{5}{2}\right) = 2$

$f(2) = 1$

$f(1) = 1$

$f(2) = 1$

$f(1) = 1$

16.(B) $\lim_{x \to \pi/3} \dfrac{-\cos\left(\dfrac{\pi}{3} - x\right)}{-2\sin x} = \dfrac{-1}{-\sqrt{3}} = \dfrac{1}{\sqrt{3}}$

17.(C) $e^{\lim_{x\to 0}\left(\frac{a\sin bx}{x\cos x}\right)} = e^{ab}$

18.(C) $\tan\frac{x}{2}(\sec x+1) = \tan x$

$\therefore \qquad f_n(x) = \tan 2^n x$

19.(A) $\lim_{x\to 0}\frac{(2e^{\sin x}+1)\,(e^{\sin x}-1)}{x\,(x+2)\times\sin x}\times\sin x = \frac{3}{2}$

20.(C) LHL + RHL $= 0+\frac{5\,(\sin(-1))}{-1+2} = -5\sin 1$

SECTION-2

21.(2) $f(x) = \begin{cases} x^2 & , -1<x<0 \\ 1 & , x=0 \\ \frac{1}{x^2} & , 0<x<1 \end{cases}$

$\therefore \lim_{x\to 0^-} f(x) + \lim_{x\to 1^-} f(x) + \lim_{x\to -1^+} f(x) = 0+1+1 = 2$

22.(3) $\lambda = \lim_{x\to 3^-}\frac{(27)^{\frac{(x+3)x}{27}}-9}{3^x-27} = \lim_{x\to 3^-}\frac{3^2\left[3^{\frac{x^2+3x}{9}-2}-1\right]}{3^3(3^{x-3}-1)}$

$= \lim_{x\to 3^-}\frac{1}{3}\left[\frac{x^2+3x-18}{9(x-3)}\right] = \lim_{x\to 3^-}\frac{1}{3}\left[\frac{2x+3}{9}\right] = \frac{1}{3}$

23.(6) $\because \qquad f(x) \to$ even + periodic

$f(-4) = f(4) = 40$

$f(-13) = f(13) = f(3) = 19$

$f(-11) = f(11) = f(1) = 2$

24.(2) $u_n = \frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+....+\frac{n}{2^n}$...[AGP]

$\frac{1}{2}u_n = \qquad \frac{1}{2^2}+\frac{2}{2^3}+....+\frac{n-1}{2^n}+\frac{n}{2^{n+1}}$

$\Rightarrow \qquad \frac{1}{2}u_n = \left(\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+....+\frac{1}{2^n}\right)-\frac{n}{2^{n+1}}$

$\Rightarrow \qquad u_n = 2\left(1-\frac{1}{2^n}\right)-\frac{n}{2^n}$

25.(1) $\lim_{x\to\frac{\pi}{4}}(1+[x])^{\frac{1}{\ln(\tan x)}} = 1^\infty = 1$